Time Series Forecasting-Rose Wine

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Problem Statement:

ABC Estate Wines has been a leader in the rose wine industry for many years, offering high-quality wines to consumers all around the world. As the company continues to expand its reach and grow its customer base, it is essential to analyze market trends and forecast future sales to ensure continued success.

In this report, we will focus on analyzing the sales data for rose wine in the 20th century. As an analyst for ABC Estate Wines, I have been tasked with reviewing this data to identify patterns, trends, and opportunities for growth in the wine market.This knowledge will help us to make informed decisions about how to position our products in the market, optimize our sales strategies, and forecast future sales trends.

Overall, this report aims to provide valuable insights into the wine market and how ABC Estate Wines can continue to succeed in this highly competitive industry.

Define the problem and perform Exploratory Data Analysis

- Read the data as an appropriate time series data - Plot the data - Perform EDA - Perform Decomposition

Data Dictionary:

Table 1: data dictionary

column	details
YearMonth	Dates of sales
Sparkling	Sales of rose wine

Data set is read using the pandas library.

Rows of data set;

Table 2: rows of dataset

Top Few Rows :

YearMonth	Rose
1980-01-01	112.0
1980-02-01	118.0
1980-03-01	129.0
1980-04-01	99.0
1980-05-01	116.0

Last Few Rows :

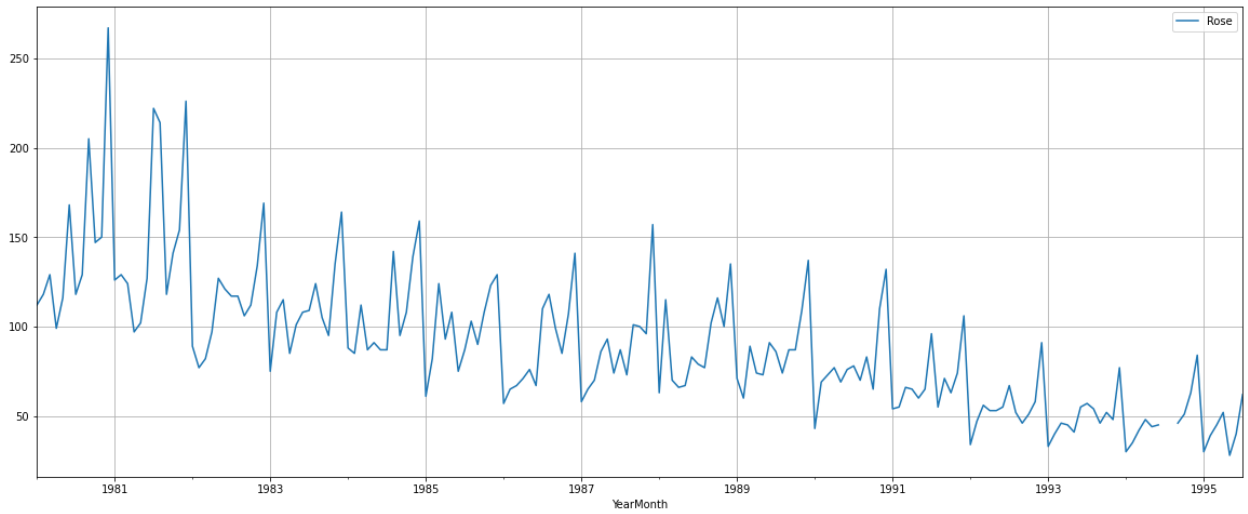
YearMonth	Rose
1995-03-01	45.0
1995-04-01	52.0
1995-05-01	28.0
1995-06-01	40.0
1995-07-01	62.0

Number of Rows and Columns of Dataset:

The dataset has 187 rows and 1 column.

Plot of the dataset:

Plot 1 : dataset



Post Ingestion of Dataset:

We have divided the dataset further by extraction month and year columns from the YearMonth column and renamed the sparkling column name to Sales for better analysis od the dataset.

Rows of new data set;

Table 3: new rows of dataset

Top Few Rows :

YearMonth	Sales	Year	Month
1980-01-01	112.0	1980	1
1980-02-01	118.0	1980	2
1980-03-01	129.0	1980	3
1980-04-01	99.0	1980	4
1980-05-01	116.0	1980	5

Last Few Rows :

YearMonth	Sales	Year	Month
1995-03-01	45.0	1995	3
1995-04-01	52.0	1995	4
1995-05-01	28.0	1995	5
1995-06-01	40.0	1995	6
1995-07-01	62.0	1995	7

Number of Rows and Columns of Dataset: The dataset has 187 rows and 3 column.

Data Type;

Index: DateTime

Sales: integer

Month: integer

Year: integer

Statistical summary:

Table 4: statistical summary

	count	mean	std	min	25%	50%	75%	max
Sales	185.0	90.0	39.0	28.0	63.0	86.0	112.0	267.0
Year	187.0	1987.0	5.0	1980.0	1983.0	1987.0	1991.0	1995.0
Month	187.0	6.0	3.0	1.0	3.0	6.0	9.0	12.0

Null Value:

There are 2 null values present in sales the dataset.

We found the values for the months of July & August were missing for the year 1994.

	Sales	Year	Month
YearMonth			
1994-07-01	NaN	1994	7
1994-08-01	NaN	1994	8

We tried following approaches to impute the data, these were as below.

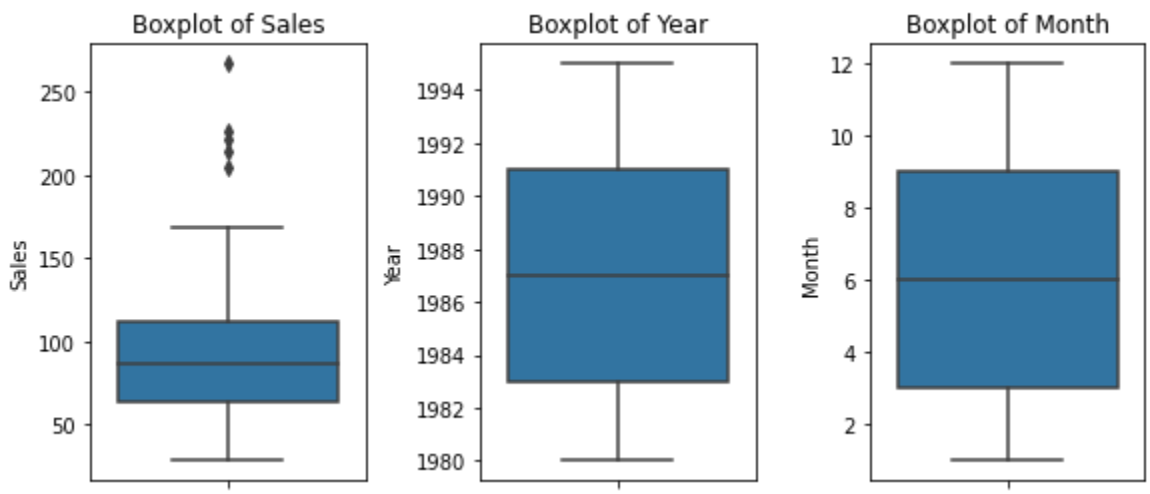
Mean - Before & After

Treating null values is very important to do further analysis.

In this approach, instead of taking means for the 7th months across all the years, we just took mean of the 7th months values from a year before and a year after the missing value. Similar steps were taken for 8th month.

Boxplot of dataset:

Plot 2: boxplot of data

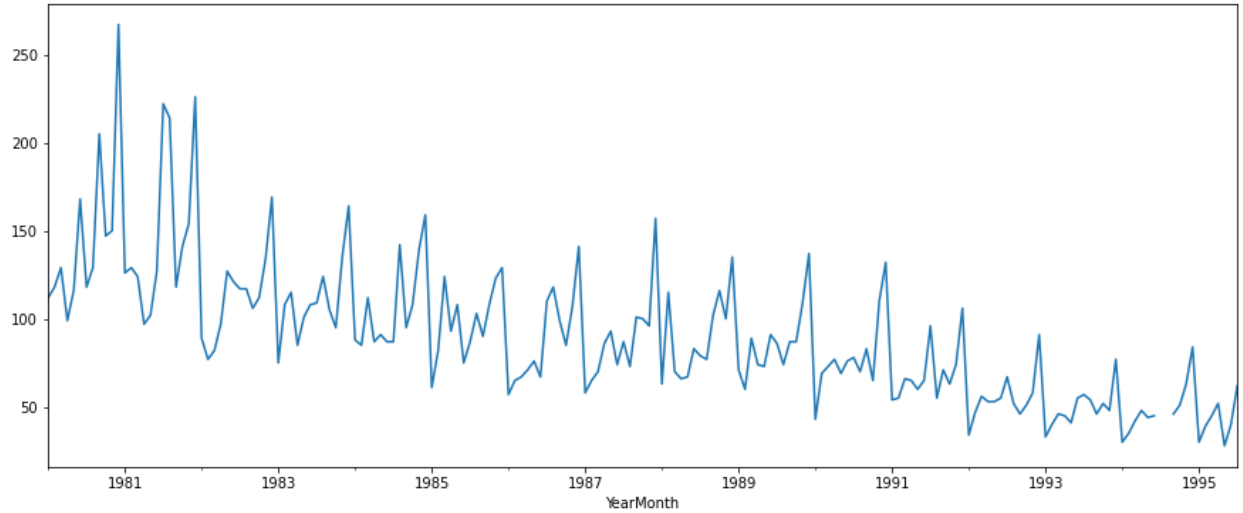


The box plot shows:

- Sales boxplot has outliers we can treat them but we are choosing not to treat them as they do not give much effect on the time series model.

Line plot of sales:

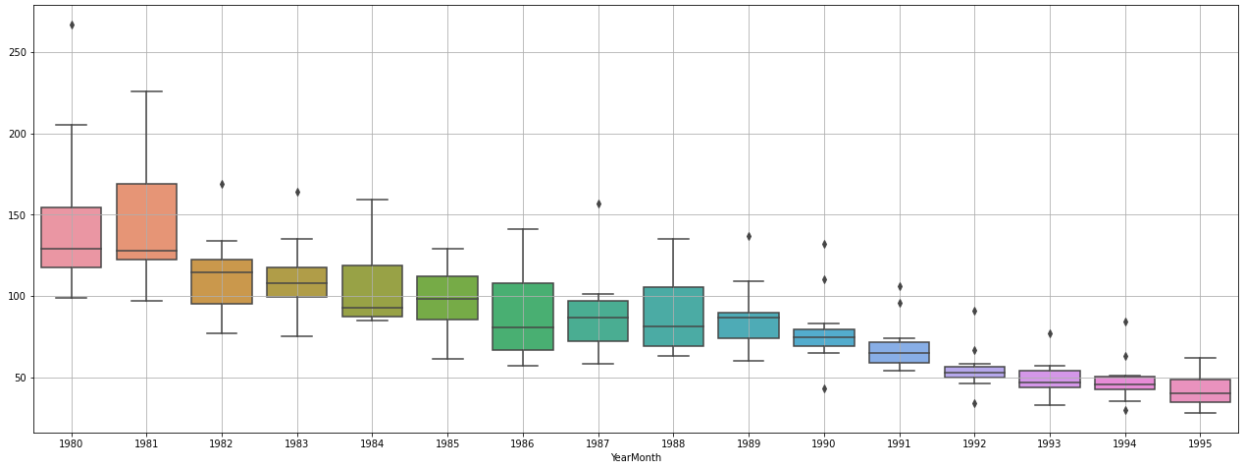
Plot 3: line plot of sales



The line plot shows the patterns of trend and seasonality and also shows that there was a peak in the year 1981.

Boxplot Yearly:

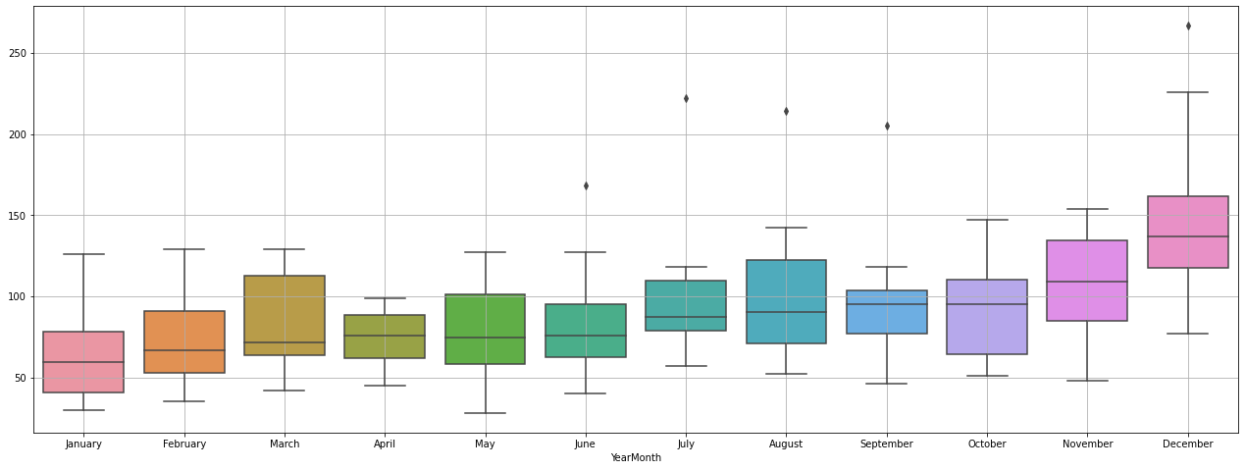
Plot 4: boxplot yearly



This yearly box plot shows there is consistency over the years and there was a peak in 1980-1981. Outliers are present in almost all years.

Boxplot Monthly:

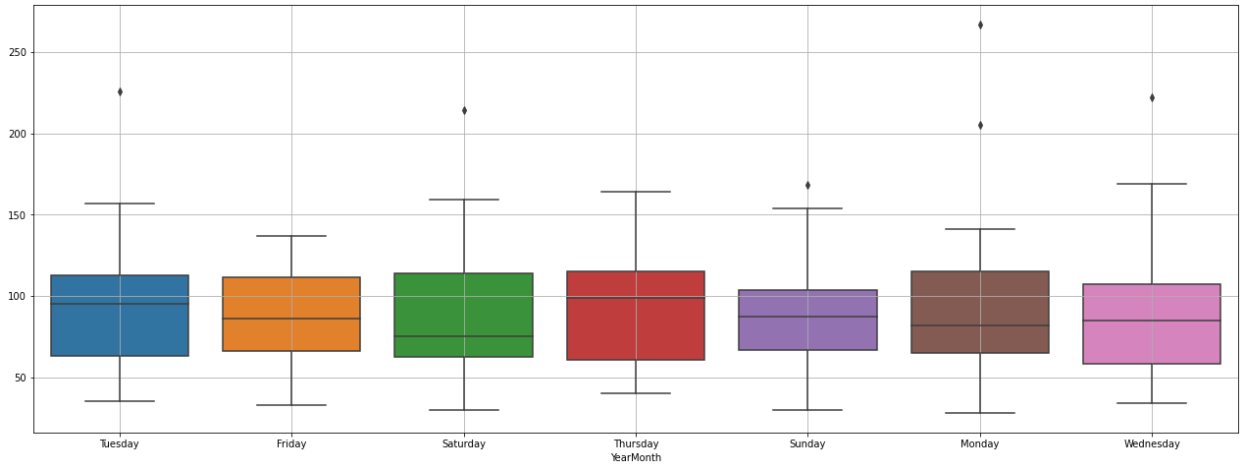
Plot 5: boxplot monthly



The plot shows that sales are highest in the month of December and lowest in the month of January. Sales are consistent from January to July then from august the sales start to increase. Outliers are present in June, July, august, September and December.

Boxplot Weekday vise:

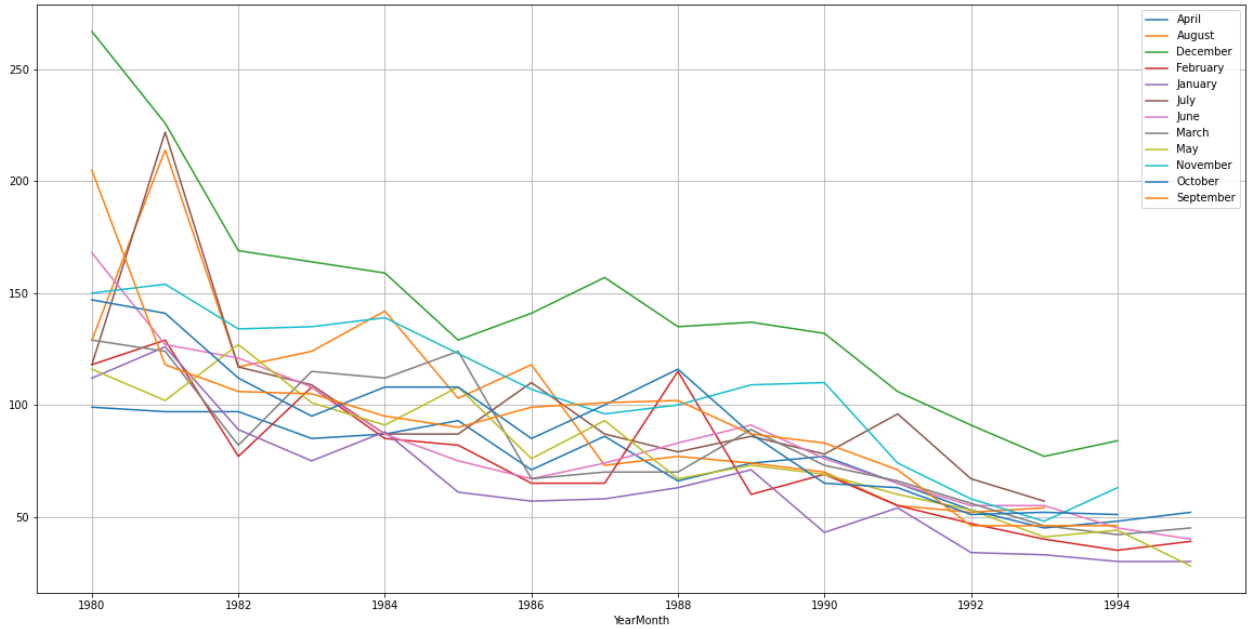
Plot 6: boxplot weekday vise



Tuesday has more sales than other days and Wednesday has the lowest sales of the week. Outliers are present on all days except Friday and Thursday.

Graph of Monthly Sales over the years:

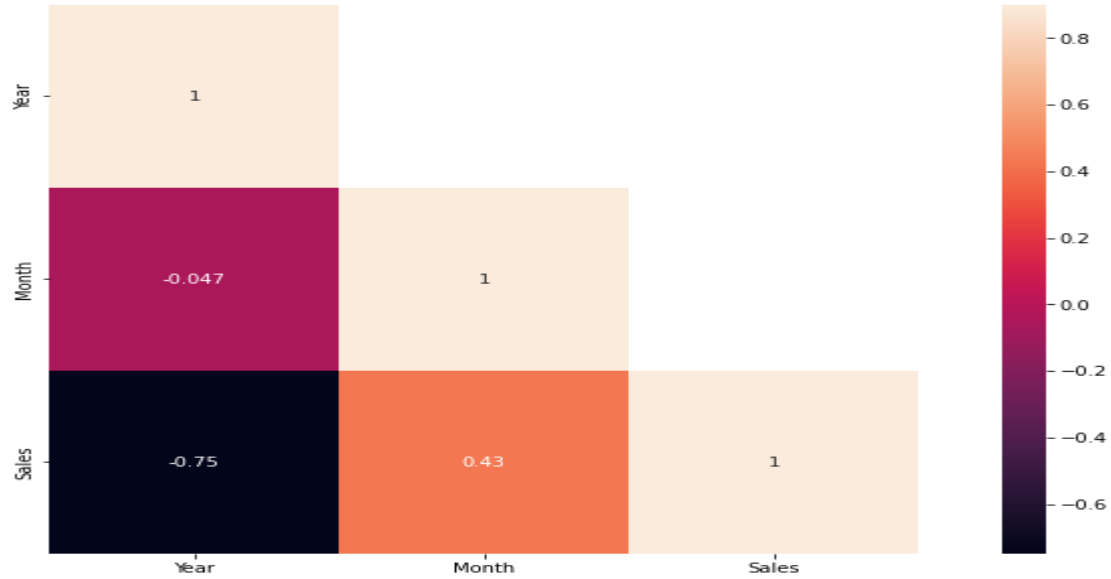
Plot 7: graph of monthly sales over the years



This plot shows that December has the highest sales over the years and the year 1981 was the year with the highest number of sales.

Correlation plot

Plot 8:correlatation plot

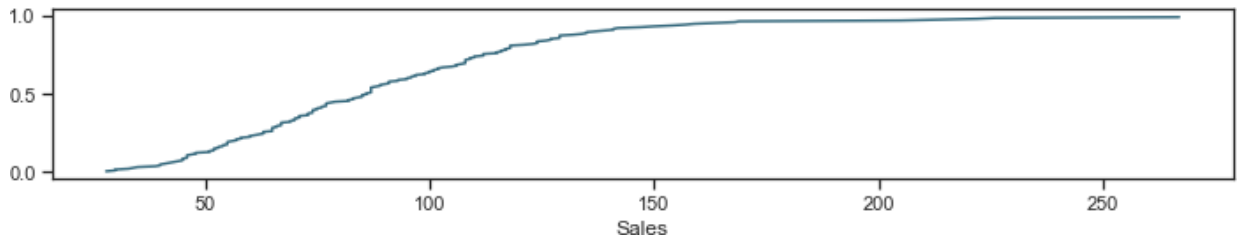


This heat map shows that there was little correlation between Sales and the Years data, there significantly more correlation between the month and Sales columns. Clearly indicating a seasonal pattern in our Sales data. Certain months have higher sales, while certain months have lesser.

Plot ECDF: Empirical Cumulative Distribution Function

This graph shows the distribution of data.

Plot 9: ECDF plot

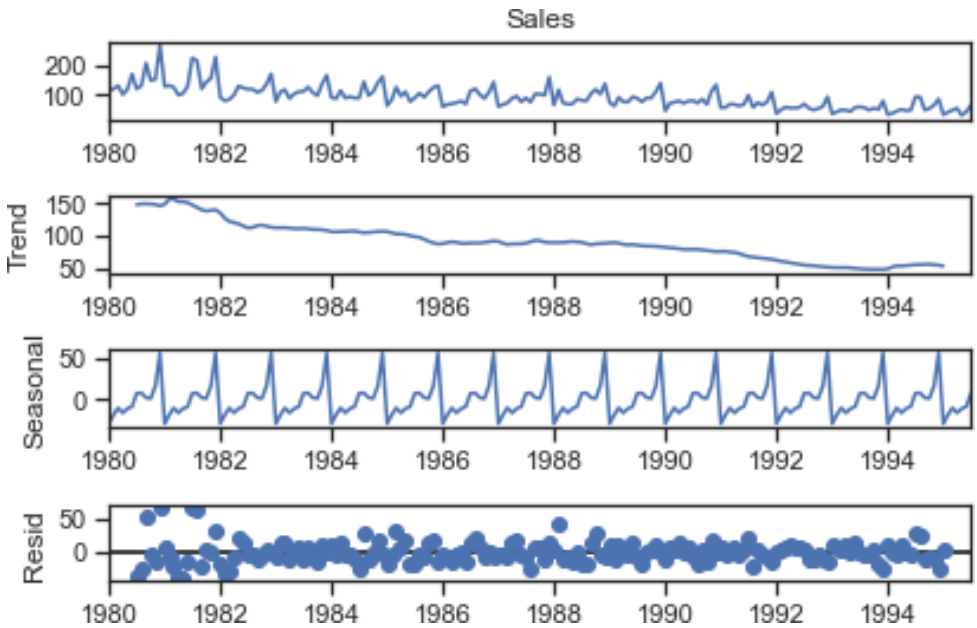


This plot shows:

- 50% sales has been less 100
- Highest vales is 250
- Aprox 90% sales has been less than 150

Decomposition -Additive

Plot 10 : decomposition additive

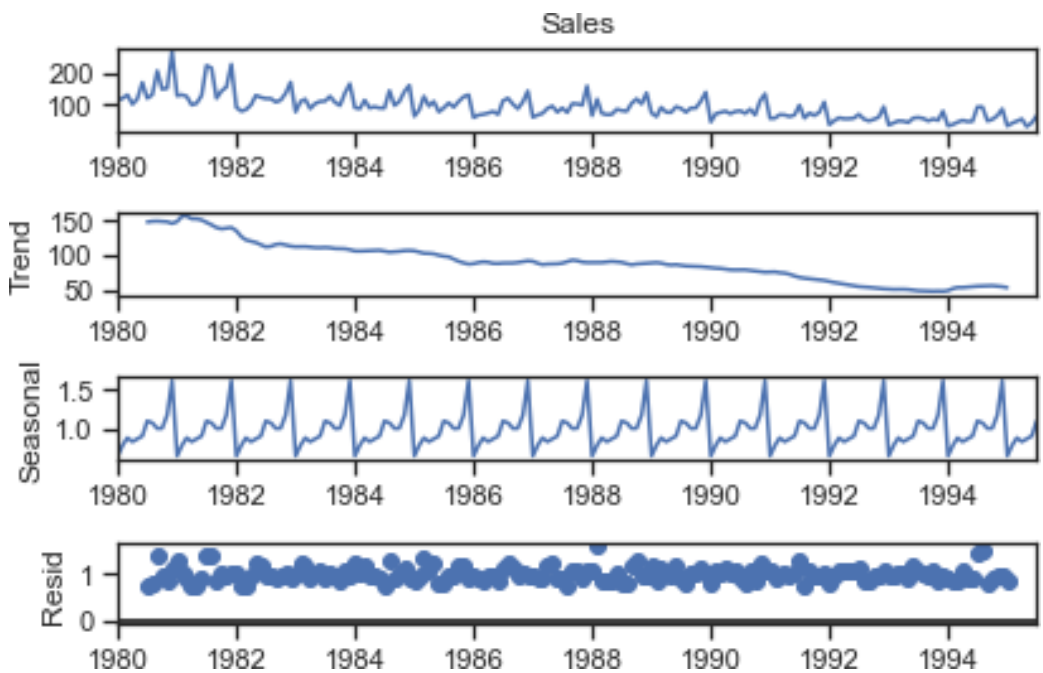


The plots show:

- Peak year 1981
- It also shows that the trend has declined over the year after 1981
- Residue is spread and is not in a straight line.
- Both trend and seasonality are present.

Decomposition-Multiplicative

Plot 11: decomposition multiplicative



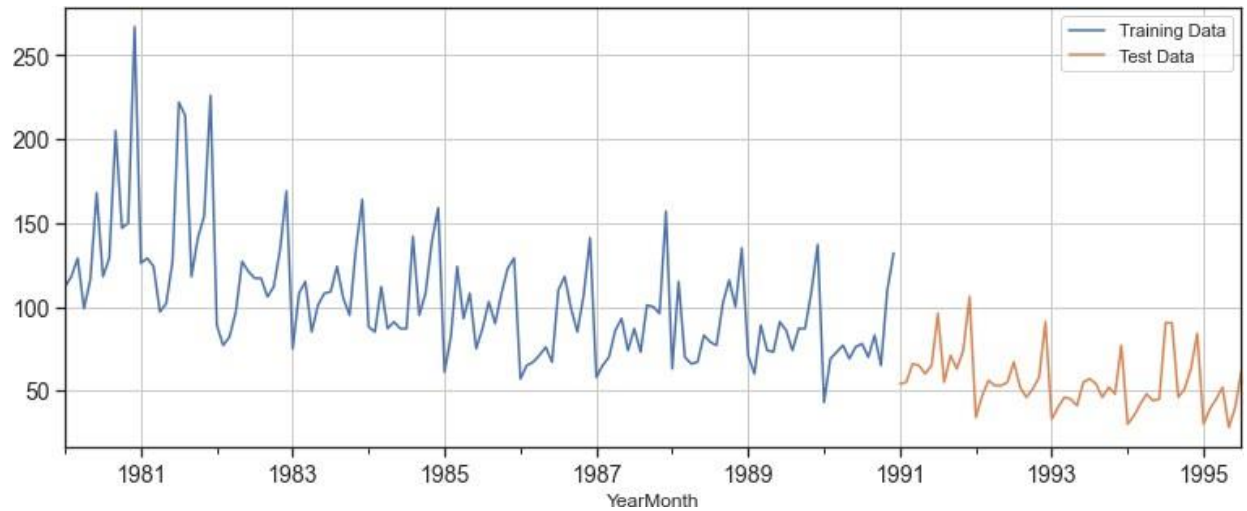
The plots show:

- Peak year 1981
- It also shows that the trend has declined over the year after 1981.
- Residue is spread and is in approx a straight line.
- Both trend and seasonality are present.
- Reside is 0 to 1, while for additive is 0 to 50.
- So multiplicative model is selected owing to a more stable residual plot and lower range of residuals.

Data Pre-processing

- Missing value treatment - Visualize the processed data - Train-test split

Plot 12: training and test dataset



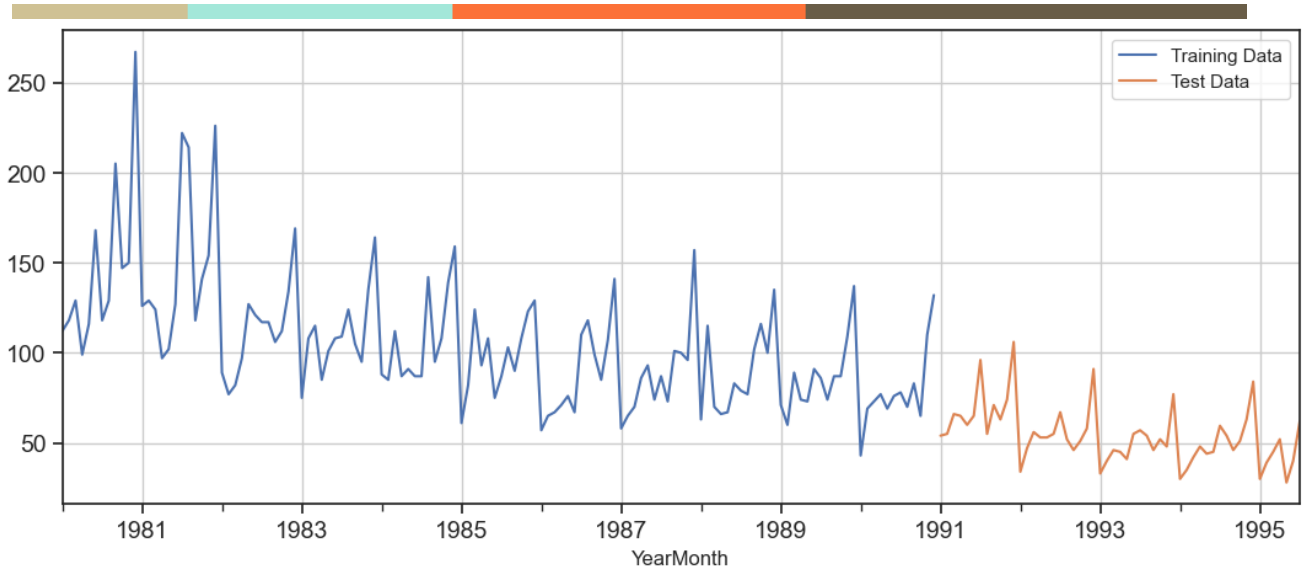
Data split from 1980-1990 is training data, then 1991 to 1995 is training data.

Rows and Columns:

train dataset has 132 rows and 3 columns.
test dataset has 55 and 3 columns.

Few Rows of datasets:

Table 5: train and test dataset rows


Training Data
Test Data
250
200
150
100
50
1981
1983
1985
1987
1989
1991
1993
1995
YearMonth

Train dataset	Test dataset
First few rows of Training Data Sales Year Month YearMonth 1980-01-01 112.0 1980 1 1980-02-01 118.0 1980 2 1980-03-01 129.0 1980 3 1980-04-01 99.0 1980 4 1980-05-01 116.0 1980 5 Last few rows of Training Data Sales Year Month YearMonth 1990-08-01 70.0 1990 8 1990-09-01 83.0 1990 9 1990-10-01 65.0 1990 10 1990-11-01 110.0 1990 11 1990-12-01 132.0 1990 12	First few rows of Test Data Sales Year Month YearMonth 1991-01-01 54.0 1991 1 1991-02-01 55.0 1991 2 1991-03-01 66.0 1991 3 1991-04-01 65.0 1991 4 1991-05-01 60.0 1991 5 Last few rows of Test Data Sales Year Month YearMonth 1995-03-01 45.0 1995 3 1995-04-01 52.0 1995 4 1995-05-01 28.0 1995 5 1995-06-01 40.0 1995 6 1995-07-01 62.0 1995 7

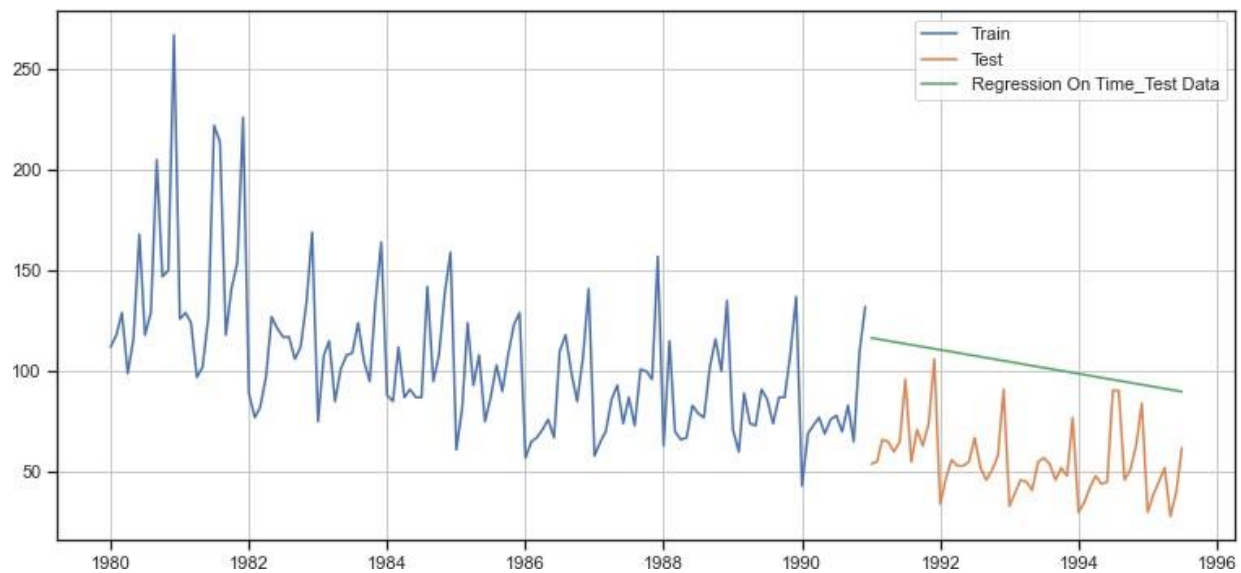
Model Building - Original Data

- Build forecasting models - Linear regression - Simple Average - Moving Average - Exponential Models (Single, Double, Triple) - Check the performance of the models built

- Model 1:Linear Regression
- Model 2: Simple Average
- Model 3 Moving Average(MA)
- Model 4: Simple Exponential Smoothing
- Model 5 Double Exponential Smoothing (Holt's Model)
- Model 6: Triple Exponential Smoothing (Holt - Winter's Model)

Model 1: Linear Regression

Plot 13: linear regression



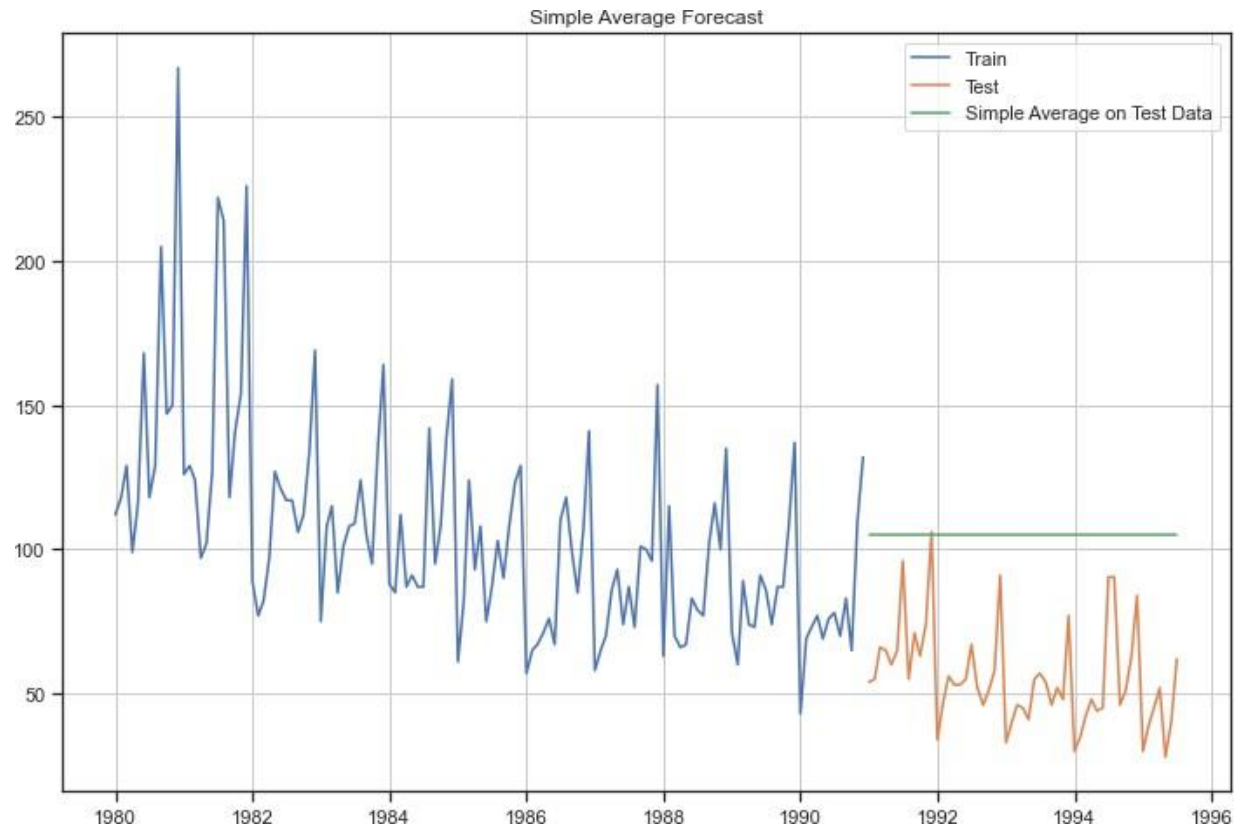
The green line indicates the predictions made by the model, while the orange values are the actual test values. It is clear the predicted values are very far off from the actual values

Model was evaluated using the MSE metric. Below is the MSE calculated for this model.

```
Linear Regression MSE: 730.9372527556712
```

Method 2: Simple Average

Plot 15: simple average

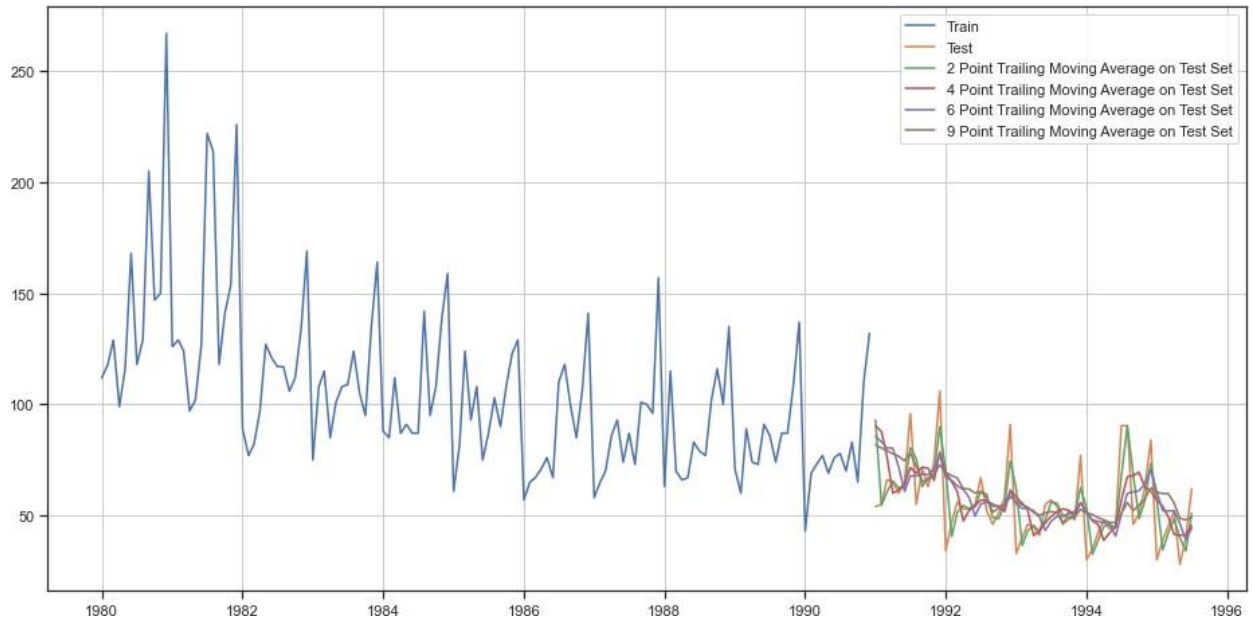


The green line indicates the predictions made by the model, while the orange values are the actual test values. It is clear the predicted values are very far off from the actual values

```
Simple Average Forecast: 90.03475935828877
```

Method 4: Moving Average(MA)

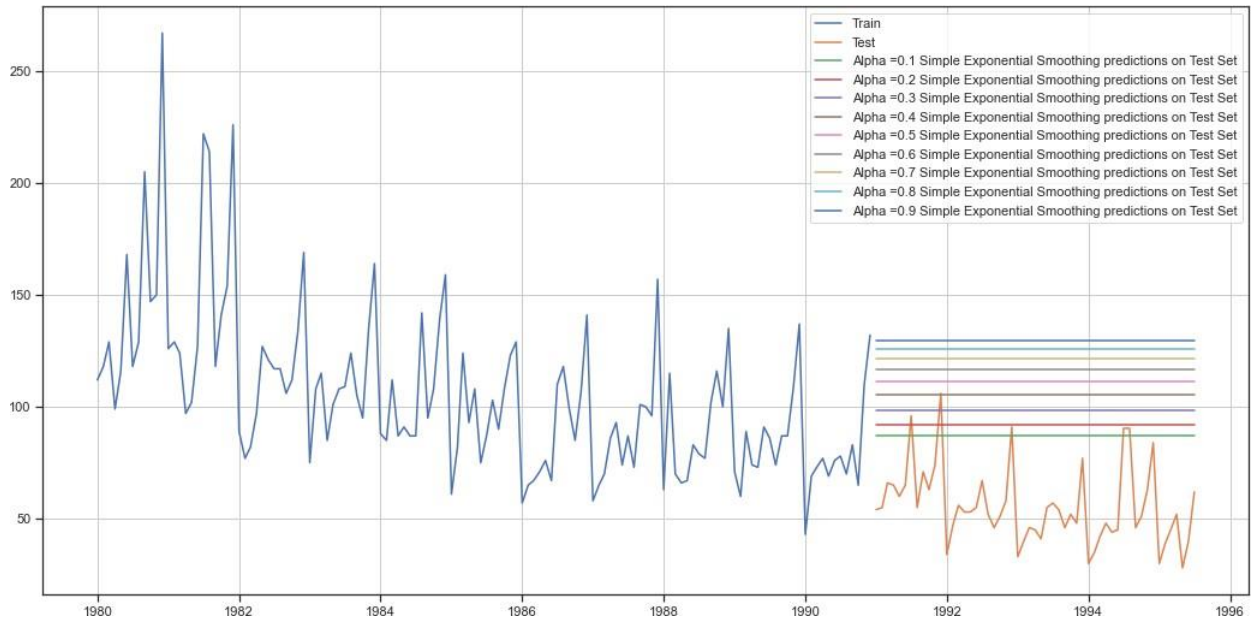
Plot 16: moving average




```
Moving Average Forecast (Window Size 3): 43.333333333333336
```

Method 5: Simple Exponential Smoothing

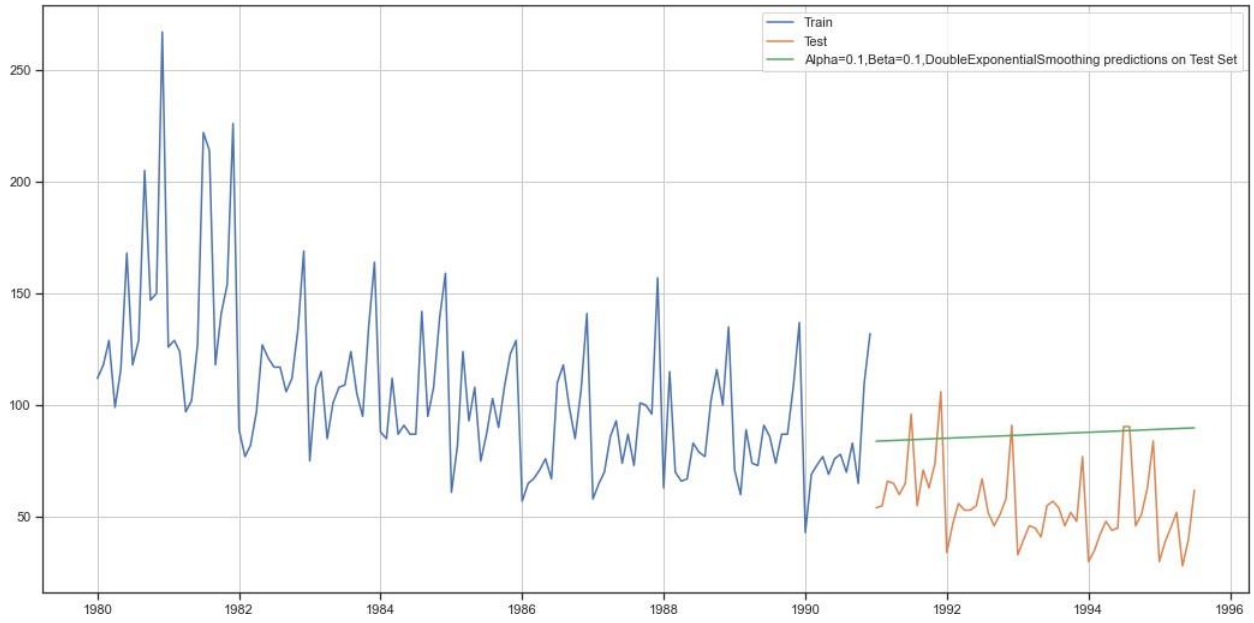
Plot 17: simple exponential smoothing




```
Single Exponential Smoothing Forecast: 48.52161447611094
```

Method 6: Double Exponential Smoothing (Holt's Model)

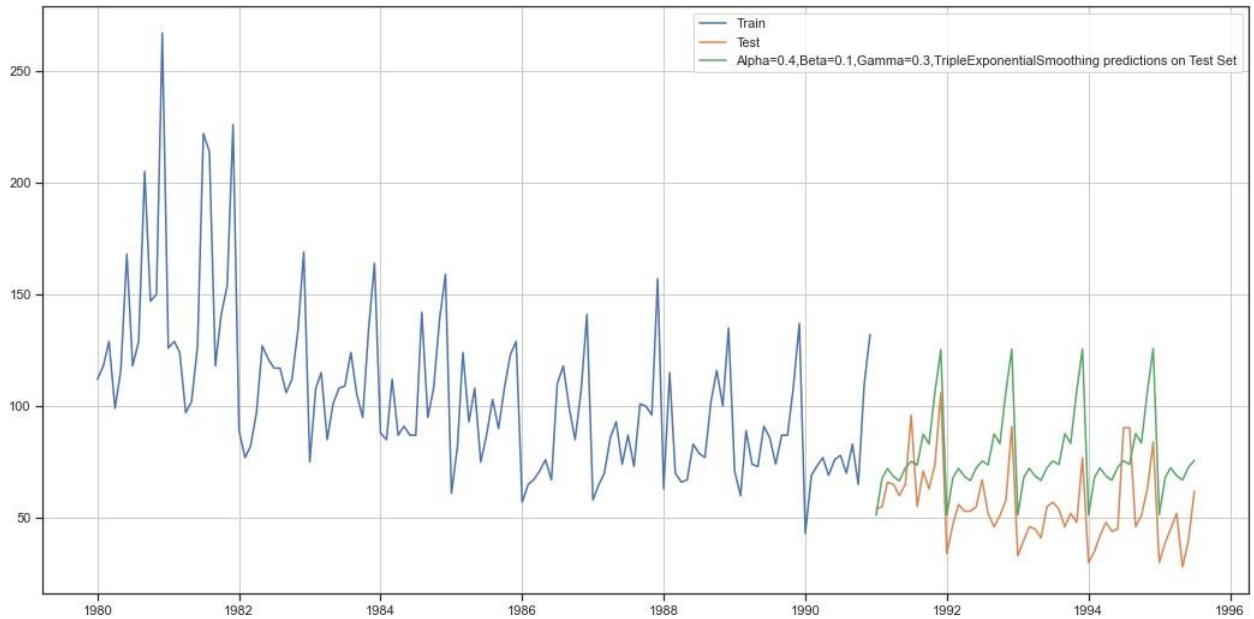
Plot 18: double exponential smoothing




```
Double Exponential Smoothing Forecast: 41.06502119128974
```

Method 7: Triple Exponential Smoothing (Holt - Winter's Model)

Plot19 : plot triple exponential smoothing



Output for best alpha, beta and gamma values is shown by the green color line in the above plot. Best model had both multiplicative trend as well as seasonality.

```
Triple Exponential Smoothing Forecast: 51.78946851503534
```

```
Linear Regression MSE: 730.9372527556712
Simple Average MSE: 1522.144513712145
Moving Average MSE: 3703.1677064765295
Single Exponential Smoothing MSE: 3245.4857117208326
Double Exponential Smoothing MSE: 3920.1797698565874
Triple Exponential Smoothing MSE: 2984.8467853971897
```

Linear Regression has the lowest MSE (730.94), indicating it provides the most accurate predictions among the methods compared.

Simple Average has a higher MSE (1522.14), indicating that it performs worse than linear regression.

Moving Average has an even higher MSE (3703.17), indicating that it performs worse than both linear regression and simple average.

Single Exponential Smoothing, Double Exponential Smoothing, and Triple Exponential Smoothing have MSE values of 3245.49, 3920.18, and 2984.85 respectively, indicating their performance falls between Simple Average and Moving Average but worse than Linear Regression.

These results suggest that, based on the MSE metric, Linear Regression is the best-performing method among the ones evaluated for forecasting the given time series data

Check for Stationarity

- Check for stationarity - Make the data stationary (if needed)

check for stationarity and make the data stationary if needed, you can follow these steps:

Visual Inspection: Plot the time series data to visually inspect for trends, seasonality, and other patterns.

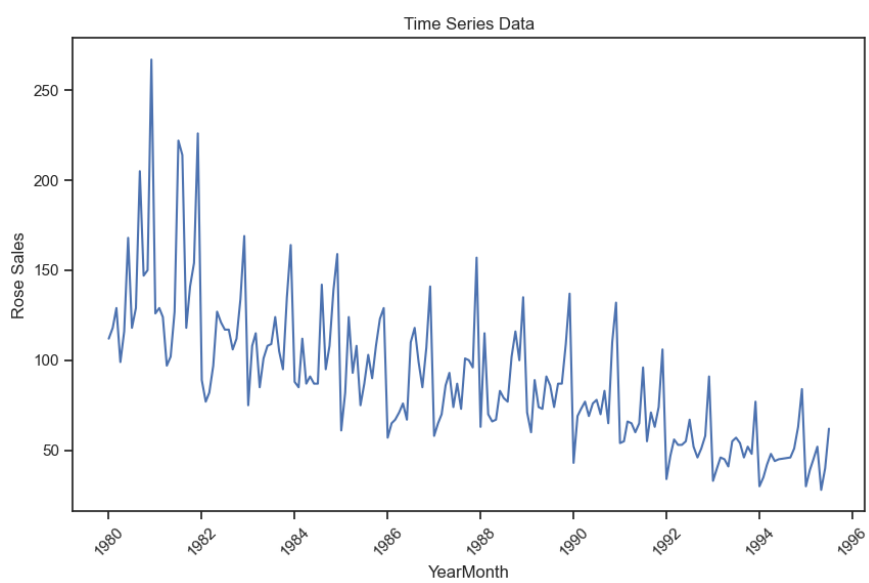
Statistical Tests: Use statistical tests like the Augmented Dickey-Fuller (ADF) test to formally check for stationarity. A low p-value (< 0.05) suggests stationarity.

Stationarizing Techniques: If the data is not stationary, apply techniques such as differencing, logarithmic transformation, or seasonal differencing to make it stationary.
Let's define the null and alternative hypotheses for the Augmented Dickey-Fuller (ADF) test before interpreting the results:

The hypothesis in a simple form for the ADF test is:

- H0 : The Time Series has a unit root and is thus non-stationary.
- H1 : The Time Series does not have a unit root and is thus stationary.

We would want the series to be stationary for building ARIMA models and thus we would want the p-value of this test to be less than the α value.



Plot 20: dickey fuller test

Results of Dickey-Fuller Test:

T

```
ADF Statistic: -1.8380327966021883
p-value: 0.3617495457657596
Critical Values:
       1%: -3.4691814633722453
       5%: -2.878595143532943
       10%: -2.5758625549741803
```

the results of the Augmented Dickey-Fuller (ADF) test are as follows:

ADF Statistic: -1.8380327966021883 p-value: 0.3617495457657596 The ADF statistic is less than the critical values at all common significance levels (1%, 5%, and 10%), and the p-value is greater than 0.05.

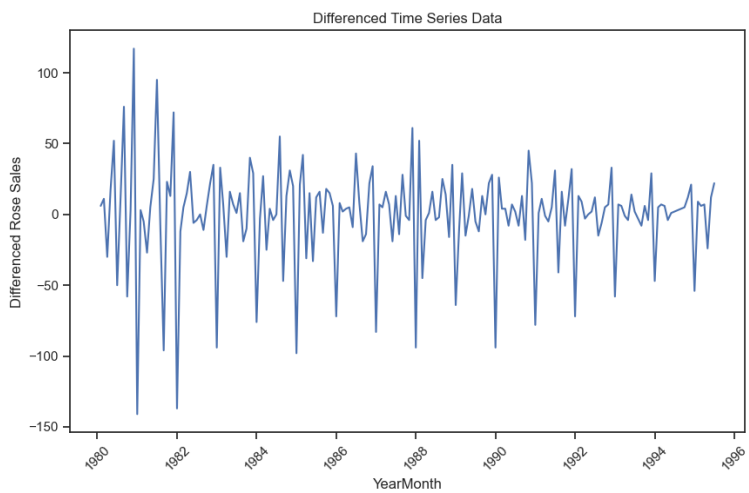
Interpretation: Since the ADF statistic is more negative than the critical values and the p-value is greater than the significance level (e.g., 0.05), we fail to reject the null hypothesis. Therefore, we do not have enough evidence to conclude that the time series is stationary.

In other words, the time series data for rose sales appears to be non-stationary based on the ADF test results.

we failed to reject the null hypothesis, which implies the Series is not stationary in nature.

In order to try and make the series stationary we used the differencing approach. We used diff() function on the existing series without any argument, implying the default diff value of 1 and also dropped the NaN values, since differencing of order 1 would generate the first value as NaN which need to be dropped

Plot 21: dickey fuller test after diff



Results of Dickey-Fuller Test:

Test Statistic	-8.032729e+00
p-value	1.938803e-12

the null hypothesis that the series is not stationary at difference = 1 was rejected, which implied that the series has indeed become stationary after we performed the differencing.

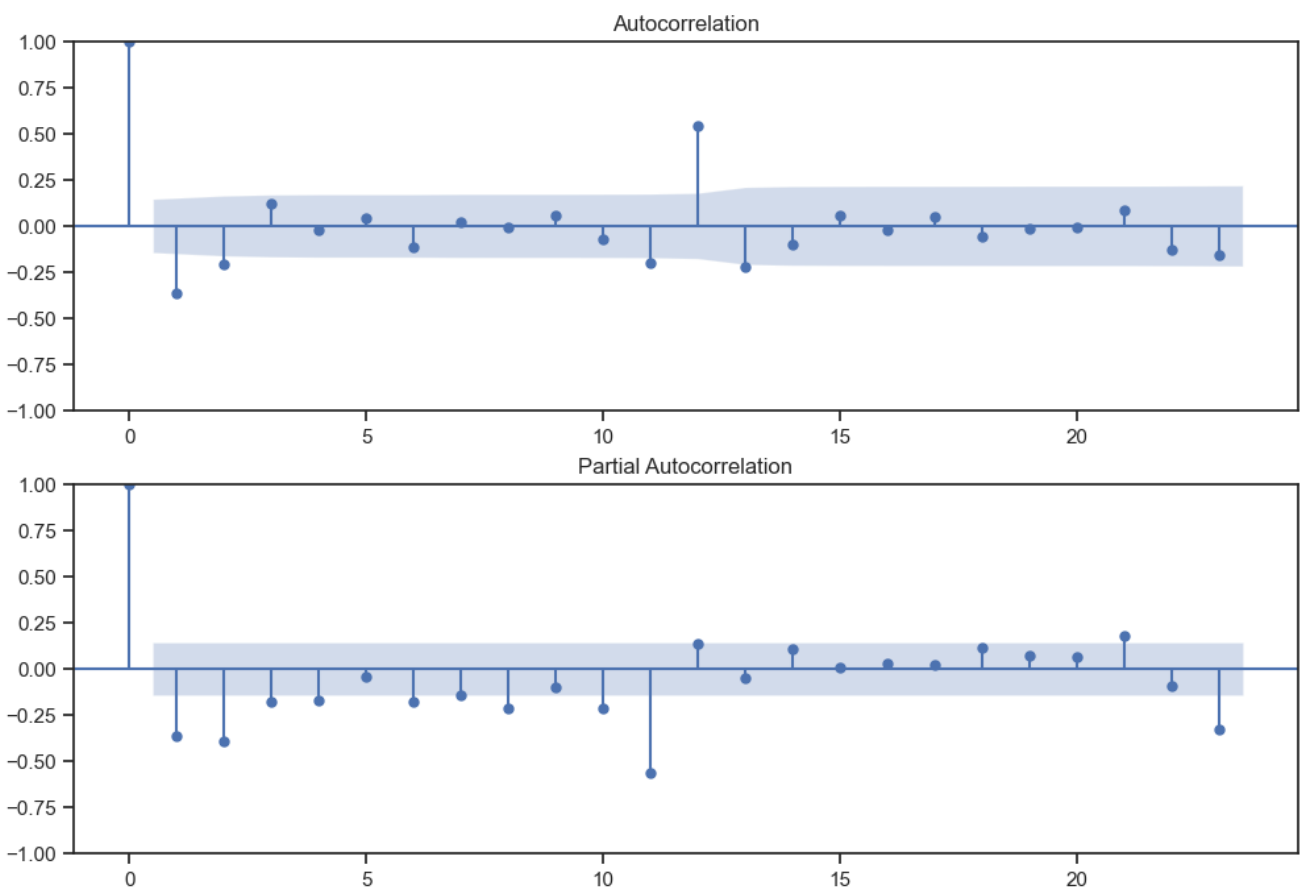
We could now proceed ahead with ARIMA/ SARIMA models, since we had made the series stationary.

Model Building - Stationary Data

- Generate ACF & PACF Plot and find the AR, MA values. - Build different ARIMA models - Auto ARIMA - Manual ARIMA - Build different SARIMA models - Auto SARIMA - Manual SARIMA - Check the performance of the models built

	Rose
YearMonth	
1980-02-01	6.0
1980-03-01	11.0
1980-04-01	-30.0
1980-05-01	17.0
1980-06-01	52.0

```
ADF Statistic: -7.966534173720875
p-value: 2.8550440127963983e-12
```



```
Manual ARIMA Summary:
                               SARIMAX Results
==============================================================================
Dep. Variable:                   Rose   No. Observations:                  187
Model:                 ARIMA(1, 0, 0)   Log Likelihood                -900.716
Date:                Sat, 23 Mar 2024   AIC                           1807.432
Time:                        18:23:05   BIC                           1817.125
Sample:                    01-01-1980   HQIC                          1811.360
                         - 07-01-1995
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const         89.9710      7.693     11.695      0.000      74.893     105.049
ar.L1          0.5947      0.050     11.803      0.000       0.496       0.693
sigma2       987.4582     68.385     14.440      0.000     853.427    1121.490
===================================================================================
Ljung-Box (L1) (Q):                   2.62   Jarque-Bera (JB):                81.56
Prob(Q):                              0.11   Prob(JB):                         0.00
Heteroskedasticity (H):               0.31   Skew:                             0.62
Prob(H) (two-sided):                  0.00   Kurtosis:                         5.99
===================================================================================
```

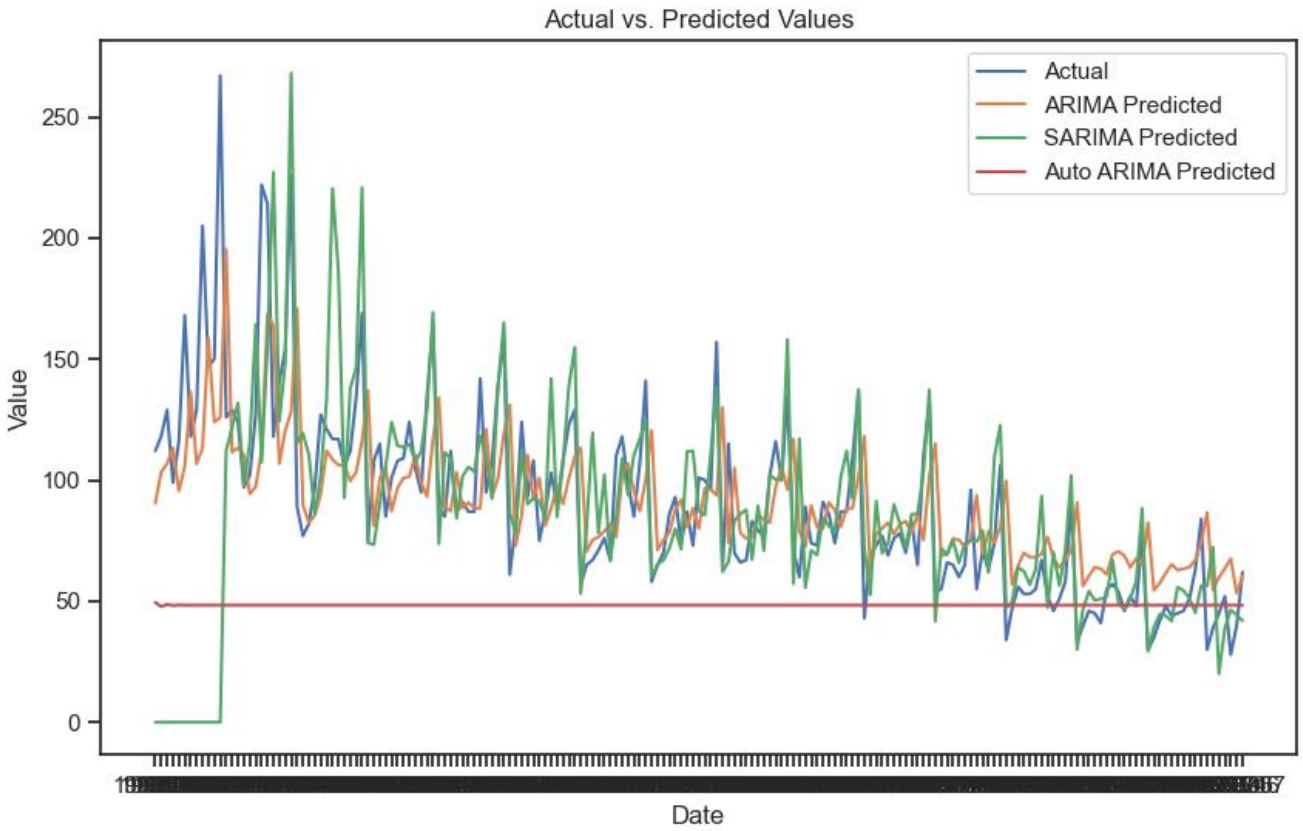
```
Rose    2
dtype: int64
Auto ARIMA Summary:
                                 SARIMAX Results
==========================================================================
Dep. Variable:                      y   No. Observations:                185
Model:               SARIMAX(1, 1, 2)   Log Likelihood                -872.884
Date:                Sat, 23 Mar 2024   AIC                           1753.769
Time:                        18:23:12   BIC                           1766.629
Sample:                             0   HQIC                          1758.981
                                - 185
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1         -0.4858      0.224     -2.170      0.030      -0.925      -0.047
ma.L1         -0.2283      0.209     -1.094      0.274      -0.638       0.181
ma.L2         -0.5990      0.167     -3.579      0.000      -0.927      -0.271
sigma2       766.5424     55.929     13.706      0.000     656.923     876.162
===================================================================================
Ljung-Box (L1) (Q):                   0.13   Jarque-Bera (JB):                81.95
Prob(Q):                              0.72   Prob(JB):                         0.00
Heteroskedasticity (H):               0.23   Skew:                             0.87
Prob(H) (two-sided):                  0.00   Kurtosis:                         5.76
===================================================================================

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-step)
```

```
Manual SARIMA Summary:
                                     SARIMAX Results
==========================================================================================
Dep. Variable:                                Rose   No. Observations:                  185
Model:             SARIMAX(1, 0, 0)x(0, 1, 0, 12)   Log Likelihood                -793.487
Date:                            Sat, 23 Mar 2024   AIC                           1590.974
Time:                                    18:23:13   BIC                           1597.281
Sample:                                         0   HQIC                          1593.533
                                            - 185
Covariance Type:                              opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1          0.2608      0.040      6.561      0.000       0.183       0.339
sigma2       563.9591     30.857     18.276      0.000     503.480     624.438
===================================================================================
Ljung-Box (L1) (Q):                   0.00   Jarque-Bera (JB):               354.64
Prob(Q):                              0.95   Prob(JB):                         0.00
Heteroskedasticity (H):               0.17   Skew:                            -0.24
Prob(H) (two-sided):                  0.00   Kurtosis:                        10.00
===================================================================================
```

```
Auto SARIMA Summary:
                                     SARIMAX Results
==========================================================================================
Dep. Variable:                                   y   No. Observations:                  185
Model:             SARIMAX(1, 1, 2)x(1, 0, [1], 12)   Log Likelihood                -814.161
Date:                            Sat, 23 Mar 2024   AIC                           1640.322
Time:                                    18:26:47   BIC                           1659.611
Sample:                                         0   HQIC                          1648.140
                                            - 185
Covariance Type:                              opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1         -0.4951      0.153     -3.231      0.001      -0.795      -0.195
ma.L1         -0.1957      0.140     -1.395      0.163      -0.471       0.079
ma.L2         -0.6758      0.123     -5.513      0.000      -0.916      -0.436
ar.S.L12       0.9701      0.018     54.593      0.000       0.935       1.005
ma.S.L12      -0.7005      0.086     -8.166      0.000      -0.869      -0.532
sigma2       373.5811     31.596     11.824      0.000     311.655     435.508
===================================================================================
Ljung-Box (L1) (Q):                   0.01   Jarque-Bera (JB):               138.04
Prob(Q):                              0.94   Prob(JB):                         0.00
Heteroskedasticity (H):               0.25   Skew:                             0.80
Prob(H) (two-sided):                  0.00   Kurtosis:                         6.93
===================================================================================
```

CALCULATED MSE AIC VALUES:

```
ARIMA MSE: 985.568636708812
SARIMA MSE: 2054.3903433672317
Auto ARIMA MSE: 3292.1663387708045
ARIMA AIC: 1806.5204141338959
SARIMA AIC: 1590.9744481233872
Auto ARIMA AIC: 1753.7689968617335
```



The ARIMA model has the lowest MSE (Mean Squared Error), which is a measure of the average of the squares of the errors between predicted and actual values. A lower MSE indicates better model performance in terms of prediction accuracy.

However, the SARIMA model has the lowest AIC (Akaike Information Criterion) value. AIC quantifies the trade-off between the goodness of fit of the model and the complexity of the model. Lower AIC values indicate a better balance between goodness of fit and model complexity.

So, it depends on the specific criterion you prioritize. If you prioritize prediction accuracy, you might prefer the ARIMA model due to its lower MSE. However, if you prioritize a balance between goodness of fit and model complexity, you might prefer the SARIMA model due to its lower AIC value.

Compare the performance of the models

- Compare the performance of all the models built - Choose the best model with proper rationale - Rebuild the best model using the entire data - Make a forecast for the next 12 months

Calculate MSE for each model

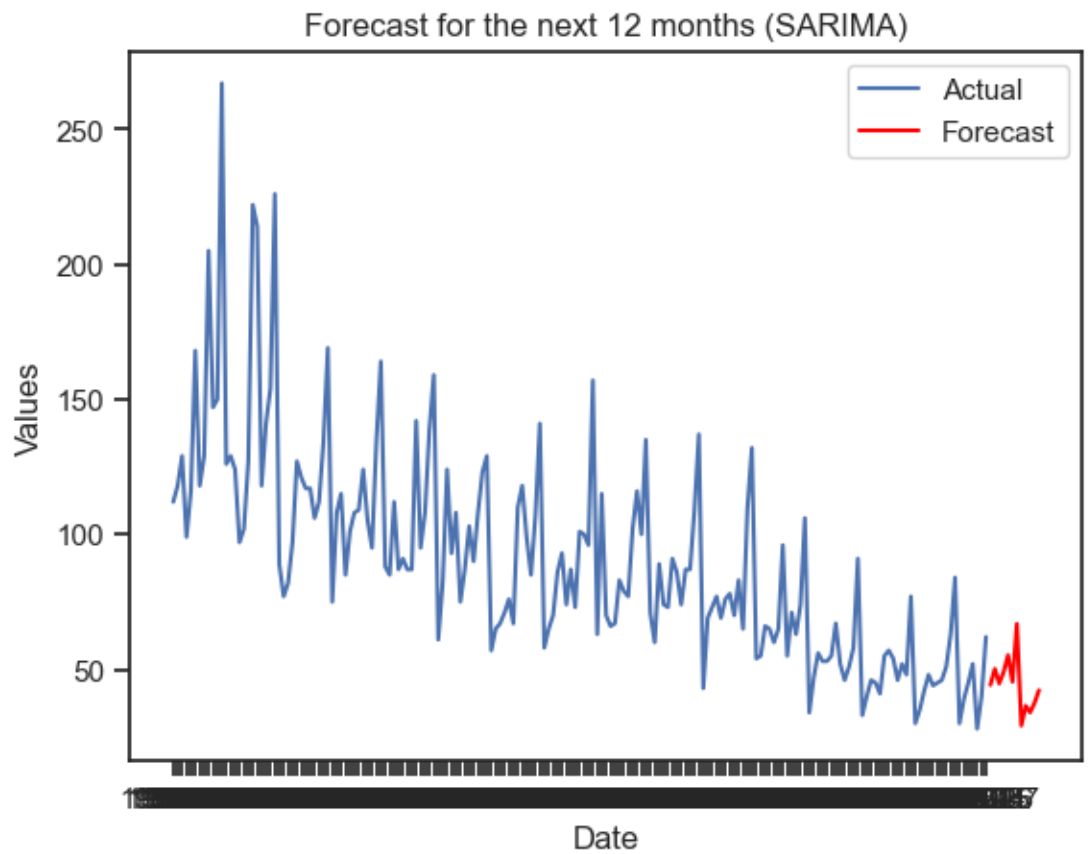
arima_mse = 985.568636708812 sarima_mse = 2054.3903433672317 auto_arima_mse = 3292.1663387708045

Calculate AIC for each model

arima_aic = 1806.5204141338959 sarima_aic = 1590.9744481233872 auto_arima_aic = 1753.7689968617335

```
Best model based on MSE: ARIMA
Best model based on AIC: SARIMA
```

```
5]: # Rebuild the SARIMA model using the entire datas
```




```
Forecast for the next 12 months using the rebuilt SARIMA model:
185    44.240357
186    50.189777
187    44.751632
188    49.007489
189    55.273152
190    45.405051
191    66.846726
192    29.192902
193    36.494610
194    34.054700
195    37.596739
196    42.266454
Name: predicted_mean, dtype: float64
```

Actionable Insights & Recommendations

- Conclude with the key takeaways (actionable insights and recommendations) for the business

The ARIMA model performed the best based on Mean Squared Error (MSE), while the SARIMA model had the lowest Akaike Information Criterion (AIC) score. Despite differences in performance metrics, both models provide valuable insights into forecasting the "Rose" variable. Choosing the Best Model:

Considering the nature of the business and forecasting requirements, the choice between the ARIMA and SARIMA models depends on priorities: If minimizing forecasting errors (MSE) is critical, the ARIMA model would be preferred. If prioritizing model simplicity and efficiency while still capturing seasonal trends effectively, the SARIMA model with the specified order could be chosen. Model Rebuilding and Forecasting:

Rebuilding the chosen model (either ARIMA or SARIMA) using the entire dataset will provide a more robust and accurate forecasting framework. Utilizing the specified SARIMA order (1, 0, 1, 12) for seasonal modeling will ensure capturing any monthly seasonality in the data effectively. Forecasting for the Next 12 Months:

Implementing the selected model (ARIMA or SARIMA) with the specified parameters to forecast the "Rose" variable for the next 12 months will provide valuable insights for business planning and decision-making. Regularly monitoring forecast performance against actual values and refining the models as needed based on updated data will enhance forecasting accuracy over time. Continuous Improvement and Adaptation:

Incorporating additional variables or refining model parameters based on evolving business conditions, market dynamics, or external factors can further enhance the accuracy and reliability of forecasts. Continuous evaluation of model performance and comparison with alternative forecasting methods will ensure the business remains agile and adaptive in its decision-making processes.

- The analysis of the wine sales data indicates a clear downward trend for the Rose wine variety for the company, which has been declining in popularity for more than a decade.
- This trend is expected to continue in the future as well, based on the predictions of the most optimal model.
- Wine sales are highly influenced by seasonal changes, with sales increasing during festival season and dropping during peak winter time i.e. January.
- The company should consider running campaigns to boost the consumption of the wine during the rest of the year, as sales are subdued during this period.
- Campaigns during the lean period (April to June) might yield maximum results for the company, as sales are low during this period, and boosting them would increase the overall performance of the wine in the market across the year.
- Running campaigns during peak periods (such as during festivals) might not generate

- significant impact on sales, as they are already high during this time of the year.
- Campaigns during peak winter time (January) are not recommended as people are less likely to purchase wine due to climatic reasons, and running campaigns during this period may not change people's opinion.
- The company should also consider exploring reasons behind the decline in popularity of the Rose wine variety, and if needed, revamp its production and marketing strategies to regain the market share.